# legio

## _travelers guide
## _for navigating the legio starsystem

**23** noise engineering firmwares · **5** from the community · **28** firmwares on one sheet

[ enc ] press · [ enc ] hold · › then · noon = 12 o'clock · cw / ccw = clockwise / counter

one 6 hp body · many minds · every control below is named by its tag on the panel drawing

labels from the noise engineering printable overlays · firmware data from each developer, 2026.10

orange = hold

## 01 the platform

lg_01 · fig.03

| | |
|---|---|
| enc | encoder + cv jack (−2 to +5 v) · some firmwares add a tap |
| 1 2 | two knobs, each with a cv jack, 0 to +5 v |
| s1 s2 | two 3-way switches, read up · mid · down |
| trig | trigger jack, about 1.6 v: hit, sync, fsu, mute… |
| in out | stereo · L feeds R when R is empty |
| LEDs | boot colours identify the firmware |
| size | 6 hp · 1.5 in deep · 140 mA +12 v · 22 mA −12 v |

## 02 swapping firmware

lg_02

1. **turn off the power to your case** and unscrew the legio (the usb port is on the back)
2. **unplug the power cable from the module**
3. plug the usb cable into the module and your computer
4. open the noise engineering firmware swap page, choose legio and a firmware, follow its steps (free, about a minute)
5. **unplug usb**, reconnect power, screw it back in · swap the panel overlay to match

needs a usb data cable and chrome, edge or opera · not firefox or safari

### community firmware

built from source

| | |
|---|---|
| flash | daisy patch sm in dfu mode (hold BOOT, tap RESET) · make program-dfu or dfu-util |
| go back | pick any noise engineering firmware on the swap page |

## 06 community

lg_06 · continued

### jawari legio

blue nautilus · github

tanpura drone: four strings strummed in turn, after the vcv module

enc tuning · push = coarse · 1 jawari shape · 2 lpf / string 1 · s1 knob 2 lpf · string · s2 formants on · off · trig strum

mono, same on both outs · pitch + semitone are saved

### rotary legio

gugenet · github

rotary speaker: horn + drum rotors, doppler, three cabinets

enc tap = max · balance · drive · 1 doppler · 2 mic distance · s1 speed max · brake · cv · s2 cab open · classic · dark · trig max speed · in in

hold the encoder 0.8 s = bypass · enc cv jack sweeps speed

## 03 oscillators + voices

lg_03 · 9 firmwares

### virt iter legio

enter virt iter sector · manual

multi-algorithm stereo oscillator · a launch firmware

enc pitch · 1 flavor · 2 tang · s1 harm · sawx · bass · s2 II · I · 0 · trig sync · in pm

### sinc legio

enter sinc sector · manual

stereo phase-modulation oscillator

enc pitch · 1 morph · 2 timbre · s1 warp · op · fold · s2 II · I · 0 · trig sync · in pm

### gator legio

enter gator sector · manual

supersaw phase-mod oscillator, stereo pm ins

enc pitch · 1 detune · 2 saw mod · s1 saw count · s2 stereoness · trig 8va · in pm

### at legio

enter at sector · manual

lfsr lo-fi oscillator after the ataraxic translatron

enc pitch · 1 tone · 2 mod · s1 mode env · rm · mix · s2 II · I · 0 · trig sync · in in

### plus legio

enter plus sector · manual

full additive voice with envelope + lfo

enc pitch · 1 time · 2 timbre · s1 mod + · off · − · s2 lfo env · on · off · trig hit · in pm

## 03 oscillators + voices

lg_03 · continued

### unda legio

enter unda sector · manual

full phase-mod voice, built for chaining

enc pitch · 1 time · 2 timbre · s1 mod +1 · 0 · −1 · s2 mode adsr · asr · ar · trig hit · in mix

### reso legio

enter reso sector · manual

karplus-strong string voice + resonator

enc pitch · 1 decay · 2 brightness · s1 body banjo · violin · guitar · s2 chorus II · I · 0 · trig hit · in excite

### marimba legio

enter marimba sector · manual

physical-model resonator voice + processor

enc pitch · 1 decay · 2 tone · s1 type tube · metal · wood · s2 mode ext · bow · hit · trig hit · in excite

### morbus legio

enter morbus sector · manual

stereo noise generator

enc frequency · 1 harsh · in: fwr · 2 color · in: clip · s1 wtf · rough · none · s2 II · I · 0 · trig mute · in in

## 04 drums

lg_04 · 7 firmwares

### kick legio

enter kick sector · manual

analog-style kick that ducks the audio at its ins

enc pitch / mute · 1 decay · 2 impact · s1 tone · s2 sat · trig hit · in duck

### snare legio

enter snare sector · manual

physical-model snare

enc pitch / mute · 1 decay · 2 noise · s1 impact · s2 wire · trig hit · in mix

### tom legio

enter tom sector · manual

toms, lasers, zaps, mid-range hits

enc pitch / mute · 1 decay · 2 punch · s1 tone · s2 sat · trig hit · in mix

### cymbal legio

enter cymbal sector · manual

granular metal for hats, rides + crashes

enc morph / mute · 1 decay · 2 tone · s1 type ride · crash · hat · s2 alloy · trig hit · in mix

### bos legio

enter bos sector · manual

more cowbell, stereo pm ins

enc pitch / mute · 1 decay · 2 tone · s1 mode fm · add · x0x · s2 tilt · trig hit · in fm

### klang legio

enter klang sector · manual

zappy percussion after basimilus

enc pitch / mute · 1 decay · 2 impact · s1 harm · s2 spread · trig hit · in mix

### tymp legio

enter tymp sector · manual

compact stereo digital drum

enc pitch · 1 decay · 2 wackitude · s1 mode boing · cat · boot · s2 tang · trig hit · in quack

## 06 community

lg_06 · 5 open firmwares

### sawstack legio

carl tashian · github · altstrument

five-voice supersaw, no envelope · also an au / vst3 plugin · mit

enc fine pitch · push = coarse · 1 detune · 2 morph · s1 mode stack · rich · sub · s2 width mono · stereo · wide · trig hard sync

### stutterer legio

carl tashian · github · altstrument

stutter › tape delay › dj filter › soft clip, clock-synced · mit

enc tape speed · tap = stutter · 1 subdivision · 2 dj filter · s1 clock ×2 · =1 · ÷2 · s2 wet 100 · 50 · 0 % · trig clock · in in

### drifter legio

carl tashian · github · altstrument

random bézier auto-panner, crossfader or cv panner · mit

enc rate · tap = curve · 1 center · 2 depth · s1 mode pan · xfade · cv pan · s2 edge clip · fold · wrap · trig clock · in in

## 05 effects

lg_05 · 7 firmwares

### librae legio

enter librae sector · manual

dynamics: compress, limit, expand, gate · a launch firmware

enc color · 1 ceiling · 2 room · s1 tilt high · off · low · s2 cut max · some · none · trig fsu · in in

### chasma legio

enter chasma sector · manual

mid / side compression + limiting

enc distortion · 1 side limit · 2 mid limit · s1 mid comp · s2 side comp · trig fsu · in in

### ampla legio

enter ampla sector · manual

multimode filter + gate, tube-style drive

enc freq · 1 a/d · 2 time · s1 howl · bite · calm · s2 hp · bp · lp · trig hit · in in

### roucha legio

enter roucha sector · manual

stereo multimode filter with folder

enc frequency · tap = bypass · 1 resonance · 2 fold · s1 all · post · pre · s2 hp · bp · lp · trig bypass · in in

### bitcrush legio

enter bitcrush sector · manual

pitch-tracking downsampler, bits + dither

enc rate / bypass · 1 bits · 2 dither · s1 env in · out · off · s2 lpf off · high · low · trig bypass · in in

### megafold legio

enter megafold sector · manual

stereo wavefolder + shaper with chorus

enc envelope · 1 gain · 2 shape · s1 chorus lots · some · none · s2 speed fast · mid · slow · trig fsu · in in

### ovis legio

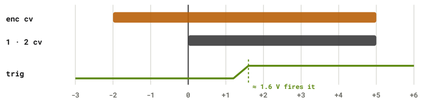
enter ovis sector · manual

granular shepard-tone processor

enc pitch · 1 count · 2 length · s1 fade · s2 layers · trig hit · in in

## __fig.05 : cv ranges

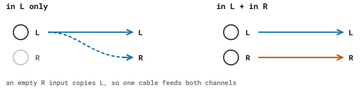
every firmware




cv adds to the control · switches have no cv

## __fig.06 : L feeds R

in · out




an empty R input copies L, so one cable feeds both channels

## __fig.03 : tagged panel

the tags every card uses

| | |
|---|---|
| [ enc ] | __encoder : turn · tap in some firmwares |
| [ 1 ] | __top knob : the first big knob |
| [ 2 ] | __bottom knob : the second big knob |
| [ s1 ] | __left switch : up · mid · down |
| [ s2 ] | __right switch : up · mid · down |
| [ trig ] | __jack : hit / sync / fsu, about 1.6 v |
| [ cv ] | __enc · 1 · 2 : a cv jack for each |
| [ in ] | __in L / R : L feeds R if alone |
| [ out ] | __out L / R : stereo out |
| [ LED ] | __two lights : boot colours = firmware |




reading a card · kick legio as an example

**kick legio** enter kick sector · manual

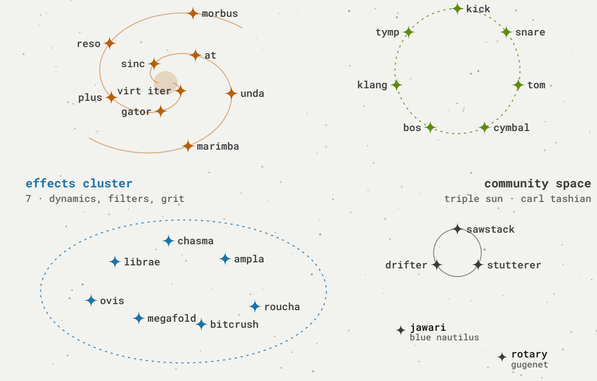
analog-style kick that ducks the audio at its ins

enc pitch / mute · 1 decay · 2 impact · s1 tone · s2 sat · trig hit · in duck

the encoder and both knobs also have a cv jack under the same name · switch positions read top to bottom · "pitch / mute" = turn for pitch, tap to mute

## __fig.04 : the star map

28 firmwares by family




rogue stars: one firmware per maker